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Teamwork

Mathematical Modeling

Adaptability

Professional Summary

Mathematical Engineering student with a strong background in analysis, probability, optimization, and an interest in the rigorous study of the mathematical foundations of machine learning. My central motivation is to contribute to the formalization of methods that today operate largely on heuristic grounds, addressing questions such as the theoretical characterization of deep learning architectures, convergence and generalization properties, and the analysis of generative models from a structured mathematical perspective.

I have practical experience in probabilistic modeling, simulation, optimization and the development of machine learning solutions in both academic settings and the financial and technology industries. I am currently taking courses from the Master's in Data Science at the University of Chile as part of a double-degree program, with the goal of graduating as a Mathematical Engineer and Master in Data Science, developing a thesis at the intersection of mathematics and machine learning.

Selected advanced coursework: Deep Generative Models, Machine Learning, Natural Language Processing, Optimal Control, Mixed-Integer Linear Programming, Markov Processes, Stochastic Calculus, Stochastic Simulation, Numerical Analysis of Partial Differential Equations.

Advanced Technical Skills

- **Languages: English** (professional working proficiency), **Mandarin Chinese** (conversational).
- **Programming Languages:** Python, R, Julia, SQL (PostgreSQL).
- **Technologies and Tools:** PyTorch, Multimodal Models, OCR, Git, PowerBI.

Work Experience and Extracurricular Activities

- Member of the **Eolian Project (Solar Car)** at the University of Chile, working mainly in the areas of motors and energy models, carrying out data analysis of electric motors for their modeling and optimization (August – December 2024).
- **Financial Risk Analyst** at **Scotiabank**, mainly responsible for the analysis, monitoring and automation of follow-ups involved in the bank's provisioning and management models (January – March 2025).
- **Machine Learning Intern at In-Data**, responsible for evaluating and comparing open-source multimodal models to automate the structured extraction of data from building permits under local hardware constraints (January – February 2026).

Selected University Projects

- **Deep Learning as an Optimal Control Problem** (Julia) — The training of deep neural networks (ResNet architecture) was formalized as a continuous-time optimal control problem, where the network depth corresponds to the time horizon and the weights are the control. The optimality conditions were derived and implemented via Pontryagin's Maximum Principle, interpreting the backpropagation algorithm as the numerical solution of the associated state-costate system of equations. The resulting classifier achieved competitive accuracy against KNN, Random Forest and SVM across four benchmark datasets.

- **Numerical Approximation of Stochastic Differential Equations via Wiener–Chaos and Fourier Neural Operators** (Python/PyTorch) — A framework was implemented combining Wiener–Chaos expansions with Fourier Neural Operators to approximate solutions of stochastic partial differential equations (SPDEs). The core idea is to decompose the stochastic solution into deterministic coefficients satisfying a coupled system of classical PDEs, allowing the application of standard numerical tools. The main case was a bistable heat equation with stochastic forcing, with preliminary exploration of 2D stochastic Navier-Stokes. A sensitivity analysis was performed with respect to the truncation order of the expansion and the hyperparameters of the neural operator.
- **Estimation of critical percolation parameters on the square grid** (Python) — Bernoulli percolation on square grids was studied through Monte Carlo simulations on grids of up to 10^4 nodes; the critical threshold $p_c = 1/2$ and the critical exponent governing the rate at which the connection probability decays near the phase transition (theoretical value $\alpha = 5/48$) were estimated. The Restart technique was applied to estimate low-probability events more optimally and accurately, along with comparing the grid’s behavior to its dual.
- **Computer vision pipeline for segmentation and volumetric counting of products on supermarket shelves** (Python) — in collaboration with **FollowUp Customer Experience**. Development of a pipeline integrating monocular depth estimation (Depth Anything V2), instance segmentation (SAM) and object detection (YOLO) to automate the issuance of preventive alerts for stockouts.
- **Implementation of Generative Models: VAE and GANs** (Python/PyTorch) — Implementation of a Variational Autoencoder and a Generative Adversarial Network for the generation and controlled modification of attributes in images, studying the structure of the learned latent space and the quality of the generated samples.
- **Stochastic optimization and mixed-integer linear programming** (Julia/JuMP) — Modeling and implementation of optimization problems under uncertainty, including two-stage formulations and mixed-integer linear programming, using the JuMP framework in Julia.

Certifications and Distinctions

- **TOEFL-ITP** certification with CEFR level C1, valid since December 2023.
- **Distinguished Student** of the Faculty of Physical and Mathematical Sciences at the University of Chile during the years 2021 and 2022.